D200 Machine Learning (ML) in Economics Project
The Lucas Critique Examined in ML

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ABSTRACT

This paper compares the performance of regime-switching ML models with classical macroeconometric models during a change in regime, using simulated data and US macroeconomic data. This paper finds that ML models which flexibly model the latent state (rather than flexibly model the immediate outcome) outperform during regime switching, in line with a theoretical framework suggested in this project.

I. INTRODUCTION

The Lucas critique [1] argues that in the estimation of a macro-econometric model, parameters of the problem are not policy invariant and, so, systematically change when a policy changes. Since models developed on a theoretical foundation are governed by agents' optimising behaviour under a particular regime, a change in regime renders a structural or reduced-form econometric model useless in constructing policy counterfactuals and forecasting. Hence, the advent of ML in providing an alternative approach to forecasting which is agnostic to the underlying structural models and may capture non-linear state dependence may improve performance during underlying regime shifts. This project therefore investigates whether ML models can exhibit greater adaptability to regime change than classical regime-switching macroeconometric models, evaluated on out-of-sample forecasting performance during periods of structural instability on simulated data and US economic data. Particularly under the rapidly switching and shifting regimes of the current macroeconomic environment, this is important for forecasting agents, including policy makers.

II. LITERATURE REVIEW

The traditional approach to macroeconometric modelling has been system estimation, underpinned by micro-founded optimal decision rules from underlying agent problems. The primitive but influential Klein model in 1950 created a system of simultaneous equations for the economy with 3 economic identities and behavioural equations, arising from theory (i.e., a Keynesian consumption function). However, in Lucas' seminal 1976 paper [1], he outlined his critique of how models based on the optimal decision rules of agents means that 'current-generation models' and 'conceivable future models' are rendered useless given a constantly changing policy regime which changes the underlying parameters in the data generating process that historical data seeks to estimate.

In the 1980s, the subsequent rise of VARs and reduced-form estimation complemented the critique [2] in aiming to construct forecasting models which were superior and more model agnostic. By simply estimating relationships between variables intertemporally without imposing strong identifying assumptions that were resulting from any particular theory but, rather, were minimal statistical impositions that were sufficient for identifying parameters, macroeconometrics saw a shift towards model agnostic estimation. Nonetheless, parameters of reduced-form econometric models are also not policy-invariant. Hence, models with time-varying parameters to reflect parameter instability have been further developed [2, 3]. By enabling regime-dependence or modelling a latent regime process, parameters can adjust more suitably in response to some regime shift, arising out of policy change, for example.

Today, much of macroeconometric modelling (i.e., DSGE) maintains many structural equations underpinned by economic theory arising from the microfounded optimal decision rules of agents [4]. Yet, since Lucas' critique, the shift towards model agnostic frameworks which are not based on structural equations has been strengthened nonetheless, firstly by VARs, parameter state dependence and, with the advent of ML. Indeed, even then, ML DSGE hybrid approaches are used in macroeconometric modelling by many instutitions [5].

This discussion is related to Box's claim that 'all models are wrong but some are useful' [6] and Breiman's two cultures [7] of data vs algorithmic modelling claim in the discipline of statistics. As noted by Athey [8], whilst a researcher in traditional economics may pick a model based on principles and estimate it once, ML allows data to select functional forms / latent states flexibly (data-driven model selection) and outperform in forecasting. In a general sense, Lucas' critique shares common principles with these concurrent developments in statistics since parameter instability can be generalised into model misspecification.

Some have begun to look at whether ML provides a resolution to the Lucas critique by estimating non-behavioural models. One paper estimates an array of ML models and compares them with traditional econometric models in their out-of-sample performance on the Progresa programme in Mexico and constructs counterfactuals for the Malawi Cash Transfer experiment, finding relative strength of ML in forecasting and the construction of counterfactuals when underlying structure may be changing [9]. Salle [10] uses an artificial neural network (ANN) expectation formation mechanism within an agent-based macroeconomic model, allowing agents to adapt mental inflation models without assuming knowledge of the underlying structural form. The ANNs continuous updates in response to new information enable expectations evolve endogenously following policy changes, thereby rendering the framework more robust to the Lucas critique and is confirmed in their results.

Nonetheless, this is an active research area and, few have considered how regime-switching/time varying parameter model equivalents in ML may perform, relative to classical regime-switching models, given the potential to capture non-linear state dependence. The pursuit of robust predictive models under regime change is highly warranted in macroeconometrics.

III. THEORY

I provide a brief illustrative theoretical overview of why data-driven methods which select models from data are likely to outperform any classical parametric econometric

approach by decomposing sources of error. Suppose the true data-generating process is regime-dependent: $y_{t+1} = f_{s_t}(x_t) + \varepsilon_{t+1}, s_t \in \{1, \ldots, S\}$, where x_t are observed macro states and s_t is a latent regime (e.g. a policy framework). Structural change implies that the mapping $f_{s_t}(\cdot)$ varies across regimes, hence, the MSE-optimal forecast is

$$m^*(x_t) = \mathbb{E}[y_{t+1} \mid x_t] = \sum_{s=1}^{S} \Pr(s_t = s \mid x_t)\, \mathbb{E}[y_{t+1} \mid x_t, s_t = s].$$

Thus forecasting requires learning both regime probabilities and within-regime conditional means.

For any forecasting model $\hat{m}(x)$, expected squared forecast risk satisfies $\mathbb{E}\big[(y_{t+1} - \hat{m}(x_t))^2\big]$

$$= \sigma^2 + \underbrace{\mathbb{E}\big[(m^*(x_t) - m(x_t; \mathcal{F}))^2\big]}_{\text{approximation error}} + \underbrace{\mathbb{E}\big[(m(x_t; \mathcal{F}) - \hat{m}(x_t))^2\big]}_{\text{estimation error}},$$

where $\mathcal{F}$ denotes the model class and $m(x_t; \mathcal{F})$ is the best model in the class. Parametric regime-switching models restrict $\mathcal{F}$ (e.g. linear f_s and low-dimensional Markov transitions), reducing estimation error but potentially inducing large approximation error when regime assignment depends richly on x_t or when f_s is nonlinear. Flexible regime-switching ML models approximate the mixture directly: $\hat{m}_{\mathrm{ML}}(x_t) = \sum_{s=1}^{S} \hat{g}_s(x_t)\, \hat{\mu}_s(x_t)$, allowing state-dependent regime probabilities $\hat{g}_s(\cdot)$ and nonlinear conditional means $\hat{\mu}_s(\cdot)$. Under the structural instability of the Lucas critique, approximation error from restrictive parametric forms can dominate estimation variance. In such environments, reducing misspecification through data-driven regime learning and model selection can lower total forecast risk and improve out-of-sample performance.

IV. DATA

Two datasets are used for this. Firstly, computational experiments are undertaken on simulated economic data where the baseline DGP is simply drawn from a two-state Markov-switching AR(1) process:

$$y_t = \mu_{s_t} + \phi_{s_t}\, y_{t-1} + \sigma_{s_t}\, \varepsilon_t, \quad \varepsilon_t \overset{\text{iid}}{\sim} \mathcal{N}(0,1), \quad P = \begin{pmatrix} 0.95 & 0.05 \\ 0.10 & 0.90 \end{pmatrix}$$

The two states, $s_t \in \{0, 1\}$, represent economic expansions (regime 0) and recessions (regime 1). The Markov transition matrix determines average regime durations with expected duration in expansion: $1/(1 - 0.95) = 20$ periods; in recession: $1/(1 - 0.90) = 10$ periods. A mild shift and severe shift are simulated to reflect policy change (i.e., gradual tightening and Volcker scale respectively) II.

Secondly, US economic data downloaded from the St Louis FRED API are used, namely, the targets are industrial production index (INDPRO) and all urban CPI (CPIAU). Effective federal funds rate (EFFR), civilian unemployment rate (UNRATE) and the binary NBER Recession indicator (USREC) are used as covariates. For the first target, INDPRO, the break date of January 1984 follows from the literature, documenting a sharp reduction in US output volatility as a break from the Global Financial Crisis [11, 12]. For the second target, the break date of October 1979 coincides with Paul Volcker's change in the Federal Reserve's operating procedure from interest-rate targeting to non-borrowed reserves targeting, reflecting a Lucas-style policy regime change also documented in literature and matched by CUSUM/Chow Tests [13] III.

V. METHODOLOGY

A variety of models are estimated/trained on pre-break data and tested on post-break data. In the case of simulated data, the RMSE reported is on a separate simulated test set of the same pre-break regime. For each model, evaluation metrics are computed, namely RMSE and a 'Lucas sensitivity ratio' (LSR) which compares pre-break RMSE and post-break RMSE which is computed with the models trained on pre-break data. Models must therefore be able to provide estimates, given that there is constant regime-switching in the training data but that the structure governing this regime switching breaks, as per the Lucas critique. Comparison enables 2 key inferences: (i) for a given model, how much does its' predictive performance degrade after a regime break and (ii) between models, which is best (and does the ML class of models outperform.

As a parametric null baseline, estimating no latent states, an AR(2) and ARMA(2,1) model are estimated. Still within the sphere of the classical regression, 2 further models are estimated: a Markov Switching Regression (MSM) and threshold autoregression (TAR). Details on the estimation of the MSM follow from the widely researched Hamilton filter which directly estimates latent states according to Markov regime switching and constructs regime-probability weighted averages. The TAR identifies regimes through a threshold variable z_t such that $s_t = \mathbf{1}[z_t > \tau], z_t = y_{t-1}$. The threshold is chosen to minimise the total sum of squared residuals across both regimes over a grid of candidates. It estimates parameters and forecasts by regime.

The next 4 models estimated all form part of the probabilistic and nonlinear ML regime models.

The Hidden Markov Model (HMM) with parameters $\theta = \{\pi, \mathbf{A}, \{\mu_k, \Sigma_k\}_{k=1}^{K}\}$ are estimated via Baum-Welch EM on the vector $\mathbf{O}_t = \Big[y_t,\ y_{t-1},\ \bar{y}_t^{(5)},\ s_t^{(5)},\ \bar{y}_t^{(20)}\Big]^\top$ where $\bar{y}_t^{(w)}$ and $s_t^{(w)}$ are the w-period trailing mean and standard deviation of y_t. Short and long run trend/volatility summaries alongside levels means regime transitions can be more reliably detected than from y_t alone. After convergence, Viterbi decoding recovers the most probable latent state sequence $s_{1:T}^*$, which partitions the training data by regime. A separate Ridge regression is then fitted on each partition. At forecast time the current regime is identified by extending the Viterbi path, giving:

$$\hat{y}_t = \hat{\beta}_{s_t^*}^\top x_t, \qquad x_t = [y_{t-1},\ \bar{y}_t^{(5)},\ s_t^{(5)},\ x_{1t},\ x_{2t}]^\top$$

The ML Regime XGB model follows a three stage pipeline using KMeans clustering on the feature space of lagged observations, covariates and trend/volatility summaries to be labelled as pseudo-regimes. Then, an XGBoost classifier reproduces the KMeans labels from the same feature set and an XGBoost regressor is trained for each regime on the labelled subset such that:

$$\hat{s}_t = \mathrm{XGBClassifier}(x_t), \qquad \hat{y}_t = \mathrm{XGBRegressor}_{\hat{s}_t}(x_t).$$

The Mixture of Experts (MoE) uses $K = 2$ Ridge regressors with a soft gating network (logistic regression) that weights them. Regime identification occurs via updating:

$$R_{tk} = P(\text{expert} = k \mid x_t, y_t) \propto \pi_k(x_t) \cdot \mathcal{N}(y_t \mid \hat{\mu}_{kt}, \hat{\sigma}_k^2).$$

$\pi_k(x_t)$ is the gating weight from the logistic regression which the EM algorithm iterates optimises by updating regime with current expert predictions and refitting each Ridge expert with sample weights to refit gating on the logistic

regression. The mixture prediction is weighted sum of expert outputs $\hat{y}_t = \sum_k \pi_k(x_t) \cdot \hat{\beta}_k^\top x_t$ and is a soft forecast: unlike TAR or XGB, each expert contributes proportionally to its gating weight rather than one model winning outright leading to partial regularisation and reduced sensitivity to regime misclassification.

Finally, the Markov Switching Neural Network (MSNN) combines the Markov chain dynamics of the MSM with per-regime MLP experts, trained jointly via EM: it replaces linear regime-specific models with neural network experts. The data-generating process is:

$$y_t = f_{s_t}(x_t) + \sigma_{s_t}\varepsilon_t, \qquad s_t \mid s_{t-1} \sim \mathbf{P},$$

where each f_k is a small feedforward MLP ($n_{\text{features}} \to 32 \to 16 \to 1$, ReLU activations, linear output) specific to regime k. Parameters are estimated via EM. The E-step runs the Hamilton forward–backward filter to compute smoothed regime posteriors $\gamma_t(k) = P(s_t = k \mid y_{1:T})$ and joint posteriors $\xi_t(j,k) = P(s_t = j, s_{t+1} = k \mid y_{1:T})$. The transition matrix is updated analytically $\hat{P}_{jk} \propto \sum_t \xi_t(j,k)$. Each network/MLP expert f_k is updated via Adam on the probability-weighted loss: $\mathcal{L}_k = \sum_t \gamma_t(k)\,(y_t - f_k(x_t))^2$. Viterbi decoding recovers the most probable state sequence $\hat{s}^*_{1:T}$, and the forecast is produced by the corresponding expert. The Markov chain constraint and smoothed posteriors act as regularisers similar to the HMM.

A model averaging ensemble of all is also created to capture a linear combination across model classes, $\mathcal{F}$.

VI. RESULTS, ROBUSTNESS & EVALUATION

The table of all results with RMSE across experiments and for all models, including the ensemble average, is in I.

TABLE I. RMSE Comparison Across All Experiments

	Simulated Mild			Simulated Severe			IP 1984			CPI 1979		
Model	Pre	Post	LSR	Pre	Post	LSR	Pre	Post	LSR	Pre	Post	LSR
AR(2) Baseline	1.634	2.706	1.66	1.634	4.261	2.61	0.717	0.514	0.72	2.631	2.594	0.99
ARMA(2,1) Baseline	1.674	5.577	3.33	1.674	11.609	6.94	0.811	0.514	0.63	2.533	3.556	1.40
Markov Switching (MSM)	1.673	5.968	3.57	1.673	11.961	7.15	0.823	0.723	0.88	2.954	4.038	1.37
Threshold (TAR)	1.537	2.526	1.64	1.537	4.266	2.78	0.624	0.397	0.64	1.912	2.276	1.19
HMM	1.696	2.559	1.51	1.696	3.892	2.29	0.648	**0.378**	0.58	2.040	2.311	1.13
Mixture of Experts	1.339	2.628	1.96	1.339	4.742	3.54	0.619	0.407	0.66	1.791	2.992	1.67
ML Regime (XGB)	0.396	3.337	8.43	0.396	8.471	21.41	0.140	0.443	3.18	0.357	2.559	7.18
MSNN	0.613	2.493	4.07	0.613	**3.391**	5.53	0.326	0.493	1.51	1.852	3.068	1.66
Model Average	1.222	**2.015**	2.47	1.222	6.007	4.92	0.521	0.401	0.77	1.757	**2.268**	1.29

LSR = post-break RMSE / pre-break RMSE. Values below 1 indicate improved forecasting performance after the break. Highlight indicates best (lowest) post-break RMSE per experiment.

Firstly, as per the simulated data and CPI 1979 data, the Lucas critique is a valid concern as in all but one case, the post-break errors are higher. The IP is unique, however. In this case, the regime shift leads to improved performance in 7/9 models (incl. ensemble average). Examining the nature of this break provides insights as this was a variance reducing break, unlike all other examined. As per our decomposition, this can be understood as the irreducible error, σ^2, falling by so much in the new regime that performance actually improves for models.

Secondly, the degradation for each given model, as implied by the LSR, does not suggest a clear link between classes of models. The ML regime XGB model has the worst degradation in all cases despite strong pre-break performance and the MSM also degrades considerably in all cases. However, a benchmark AR(2), for example, has both reasonably strong post-break performance and some of the lowest LSR scores. The models which degrades the least is the HMM which, importantly, is flexible in its latent state estimation, rather than being flexible on the functional form (like XGB) but restrictive in its latent state estimation (via K-means clustering): the latter results in overfitting and degradation, as per the Lucas critique. In fact, correct initial specification (the MSM), leads to greater degradation than flexible non-linear models.

Thirdly, out-of-sample RMSE performance between models doesn't necessarily correlate with the degradation (LSR) for a given model. Indeed, the MSNN produces the best post-break RMSE in both simulated experiments (excluding the ensemble in the mild case) despite degrading seriously.

Finally, in 2 out of the 4 cases, the model average produces a lower post-break RMSE and also produces mild LSRs in all cases. This suggests that there are gains to ensembling across model classes. The theoretical decomposition may justify this as expanding the model class $\mathcal{F}$ reduces the source of approximation error and, across many models, estimation error can also be reduced.

VII. CONCLUSIONS

ML does not explicitly resolve the Lucas critique - it can still produce better out of sample forecasts but these worsen under unforeseen (policy-induced) regime change by similar amounts. Those which outperform tend to be those which flexibly model the latent state, rather than the functional form and, perhaps contrary to expectation, correct specification can be detrimental. The degradation also depends on the nature of the regime change as, in variance reducing breaks (IP 1984), the results suggest that regime change can lead to improvements (a sort of inverse Lucas-critique).

The ensembling across many models which both tends to outperform and demonstrate greater stability provides a key insight for future research to explore. Consideration of online updating and where this most improves resilience by model memory may also prove relevant for future research but is not estimated here, where a worst case degradation is, effectively, estimated.

Appendix A: Additional tables and figures

TABLE II. Post-Break DGP: Lucas Shift Scenarios

	Mild Shift		Severe Shift	
Parameter	Pre	Post	Pre	Post
μ_0	0.5	0.2	0.5	−0.5
μ_1	−1.5	−1.0	−1.5	−3.0
σ_0	1.0	1.3	1.0	2.0
σ_1	2.5	3.0	2.5	4.5
P_{00}	0.95	0.90	0.95	0.80

TABLE III. Dataset: Break Points and Sample Periods

Dataset	Split	Period	Obs	$\bar{y}$
IP (1984)	Pre-break	1960-01 to 1983-12	268	0.291%
	Post-break	1984-01 to 2007-07	283	0.237%
CPI (1979)	Pre-break	1960-01 to 1979-09	217	5.034%
	Post-break	1979-10 to 1999-09	240	3.801%

TABLE IV. Feature Summary

Feature	Formula	Intuition
`y_lag1`	y_{t-1}	Most recent value
`y_lag2`	y_{t-2}	Two-period lag
`roll_mean_5`	$\bar{y}_t^{(5)} = \frac{1}{5}\sum_{i=0}^{4} y_{t-i}$	Short-run trend
`roll_std_5`	$s_t^{(5)}$	Short-run volatility
`roll_mean_20`	$\bar{y}_t^{(20)}$	Long-run trend
`roll_std_20`	$s_t^{(20)}$	Long-run volatility
`exog_0`	EFFR (simulated / real)	Interest rate level
`exog_1`	Simulated / UNRATE	Labour market

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